Fruit and vegetable consumption and all-cause mortality: a dose-response analysis.

BACKGROUND: The association between fruit and vegetable (FV) consumption and overall mortality has seldom been investigated in large cohort studies. Findings from the few available studies are inconsistent. OBJECTIVE: The objective was to examine the dose-response relation between FV consumption and mortality, in terms of both time and rate, in a large prospective cohort of Swedish men and women. DESIGN: FV consumption was assessed through a self-administrated questionnaire in a population-based cohort of 71,706 participants (38,221 men and 33,485 women) aged 45-83 y. We performed a dose-response analysis to evaluate 10th survival percentile differences (PDs) by using Laplace regression and estimated HRs by using Cox regression. RESULTS: During 13 y of follow-up, 11,439 deaths (6803 men and 4636 women) occurred in the cohort. In comparison with 5 servings FV/d, a lower consumption was progressively associated with shorter survival and higher mortality rates. Those who never consumed FV lived 3 y shorter (PD: -37 mo; 95% CI: -58, -16 mo) and had a 53% higher mortality rate (HR: 1.53; 95% CI: 1.19, 1.99) than did those who consumed 5 servings FV/d. Consideration of fruit and vegetables separately showed that those who never consumed fruit lived 19 mo shorter (PD: -19 mo; 95% CI: -29, -10 mo) than did those who ate 1 fruit/d. Participants who consumed 3 vegetables/d lived 32 mo longer than did those who never consumed vegetables (PD: 32 mo; 96% CI: 13, 51 mo). CONCLUSION: FV consumption <5 servings/d is associated with progressively shorter survival and higher mortality rates. The Swedish Mammography Cohort and the Cohort of Swedish Men were registered at clinicaltrials.gov as NCT01127698 and NCT01127711, respectively.